CBSE CLASS 10 | MATHEMATICS

QUADRATIC EQUATIONS

Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

Structure the equation. Inspect the discriminant.
Solve with absolute precision.

Standard Form | Factorization | Sreedharacharya Formula | Nature of Roots | Kinematics

A student-first Maths Manthan conceptual and visual revision resource



What This Handbook Includes

- Standard form criteria and parabolic geometry of roots
- Splitting middle term with rational and radical surds
- Sreedharacharya derivation of the Quadratic Formula
- Tri-state discriminant analysis ($D > 0, D = 0, D < 0$)
- High-frequency parameter k equal-roots protocols
- Complete word problem blueprints: streams, taps, work

Pedagogical Strategy: Foundations First

A quadratic equation represents the balance of squared relationships. Roots are not mere symbols; geometrically, they are the points where the trajectory crosses ground level ($y = 0$).

$$\mathbf{y = ax^2 + bx + c = 0} \iff \text{Roots on } \mathbf{x}\text{-axis}$$

$$\mathbf{D = b^2 - 4ac} \iff \text{Nature of Roots}$$

Always structure equations into standard form before applying formulas!

CBSE 2026–27 Curriculum Verification & Rationalized Scope

Official Scope (Prescribed & Tested): Standard form of a quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$). Solutions of quadratic equations (only real roots) by factorization and by using the quadratic formula. Relationship between discriminant and nature of roots. Situational word problems arising from daily activities.

Excluded / Deleted Topics (Non-Examinable):

- **"Completing the Square" Method:** Completely deleted from the NCERT syllabus.
- **Complex / Imaginary Roots:** When $D < 0$, students must simply state "No real roots exist".
- **Equations Reducible to Quadratic Form:** Complex multi-nested substitutions are deleted.

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Prepared for learning and conceptual mastery – Strictly aligned with CBSE Class 10 Syllabus 2026–27.

CHAPTER 04

CORE SYLLABUS

QUADRATIC EQUATIONS

Module 1 | Foundations: Standard Form, Degree & Physical Origins

1.1 Definition and the Critical Leading Coefficient Condition

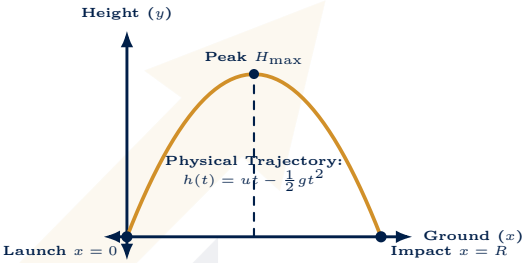
A quadratic equation in the single variable x is an algebraic equation of second degree.

The Standard Form of a Quadratic Equation:

$$\mathbf{ax^2 + bx + c = 0}$$

where $a, b, c \in \mathbb{R}$ are real numbers, and $\mathbf{a \neq 0}$.

- **Leading Coefficient (a):** The multiplier of x^2. If $a = 0$, the equation degenerates into $bx + c = 0$, which is **linear**, not quadratic!
- **Linear Coefficient (b):** Can be zero (e.g., $ax^2 + c = 0$ is a pure quadratic equation).
- **Constant Term (c):** Can be zero (e.g., $ax^2 + bx = 0 \implies x(ax + b) = 0$).



1.2 Testing Whether an Algebraic Expression is Quadratic

TRAP ALERT

Never Judge an Equation Before Full Expansion! An equation cannot be classified by visual inspection alone. You must expand all products, clear denominators, and gather all terms to one side:

- Example A: $(x-2)^2 + 1 = 2x - 3 \implies x^2 - 4x + 4 + 1 = 2x - 3 \implies \mathbf{x^2 - 6x + 8 = 0}$ **(Quadratic!)**
- Example B: $x(x+1) + 8 = (x+2)(x-2) \implies x^2 + x + 8 = x^2 - 4 \implies \mathbf{x + 12 = 0}$ **(Linear, NOT Quadratic!)**
- Example C: $x + \frac{1}{x} = x^2 \implies x^2 + 1 = x^3 \implies \mathbf{x^3 - x^2 - 1 = 0}$ **(Cubic, NOT Quadratic!)**

Worked Archetype 1 | Algebraic Simplification & Standard Form Verification

Problem: Check whether the equation $(x+2)^3 = 2x(x^2 - 1)$ is a quadratic equation. If so, write it in standard form.

Step 1: Expand LHS using algebraic identity $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$:

$$\text{LHS} = (x+2)^3 = x^3 + 3(x^2)(2) + 3(x)(2^2) + 2^3 = x^3 + 6x^2 + 12x + 8$$

Step 2: Expand RHS: $\text{RHS} = 2x(x^2 - 1) = 2x^3 - 2x$.

Step 3: Equate LHS and RHS and transfer all terms to one side:

$$x^3 + 6x^2 + 12x + 8 = 2x^3 - 2x \implies 2x^3 - x^3 - 6x^2 - 2x - 12x - 8 = 0$$

$$\mathbf{x^3 - 6x^2 - 14x - 8 = 0}$$

Conclusion: Since the highest power of x is 3 (degree is 3), this is a **cubic equation, NOT a quadratic equation**.

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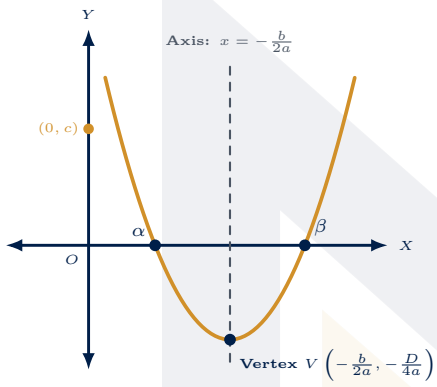
QUADRATIC EQUATIONS

Module 2 | Parabolic Geometry: Roots, Intercepts & Symmetry

2.1 Geometric Meaning of Roots: x-Intercepts of a Parabola

A real number α is called a **root** (or solution/zero) of the quadratic equation $ax^2 + bx + c = 0$ if and only if $a\alpha^2 + b\alpha + c = 0$.

- Geometrically, the roots are the **x-coordinates of the points where the parabola $y = ax^2 + bx + c$ intersects the horizontal x-axis**.
- Since a parabola can intersect a line in at most two points, a quadratic equation can have **at most two real roots**.



Anatomy of the Quadratic Function $y = ax^2 + bx + c$:

- **Curvature Direction:**
  - If $a > 0$: Opens upwards ($\cup$, concave up, has a minimum).
  - If $a < 0$: Opens downwards ($\cap$, concave down, has a maximum).
- **Axis of Symmetry:** Passes through the exact midpoint of roots:

$$\mathbf{x = \frac{\alpha + \beta}{2} = -\frac{b}{2a}}$$

- **Vertex Coordinates:** Extreme point of parabola:

$$\mathbf{V\left(-\frac{b}{2a}, -\frac{D}{4a}\right)} \quad \text{where } D = b^2 - 4ac$$

- **y-Intercept:** Set $x = 0 \implies \mathbf{y = c}$.

2.2 The Zero-Product Principle

Fundamental Algebraic Zero-Product Axiom:

$$\mathbf{A \cdot B = 0} \iff \mathbf{A = 0} \quad \text{or} \quad \mathbf{B = 0} \quad \text{(or both)}$$

If a quadratic equation is factorized into linear factors $(x - \alpha)(x - \beta) = 0$, the roots are $\mathbf{x = \alpha}$ and $\mathbf{x = \beta}$.

Worked Archetype 2 | Verifying Roots of a Quadratic Equation

Problem: Determine whether $x = -1$ and $x = \frac{1}{2}$ are solutions of the equation $2x^2 - x - 1 = 0$.

Check for $x = -1$: Substitute $x = -1$ into LHS:

$$\text{LHS} = 2(-1)^2 - (-1) - 1 = 2(1) + 1 - 1 = 2 \neq 0 \implies \mathbf{x = -1} \text{ is NOT a root.}$$

Check for $x = 1$: Substitute $x = 1$ into LHS:

$$\text{LHS} = 2(1)^2 - (1) - 1 = 2 - 1 - 1 = 0 = \text{RHS} \implies \mathbf{x = 1} \text{ IS a root.}$$

Check for $x = -\frac{1}{2}$: $\text{LHS} = 2\left(-\frac{1}{2}\right)^2 - \left(-\frac{1}{2}\right) - 1 = 2\left(\frac{1}{4}\right) + \frac{1}{2} - 1 = \frac{1}{2} + \frac{1}{2} - 1 = 0 \implies \mathbf{x = -\frac{1}{2}}$ IS a root.

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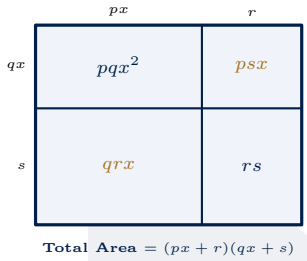
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Module 3 | Factoring by Splitting the Middle Term & Radical Surds

3.1 The AC-Method Protocol for Factoring $ax^2 + bx + c = 0$

To factor $ax^2 + bx + c = 0$, we decompose the middle term bx into two terms px and qx such that:

$$\mathbf{p + q = b} \quad \text{and} \quad \mathbf{p \cdot q = a \cdot c}$$



Step-by-Step Factoring Protocol:

1. Multiply a and c to form the product $P = ac$.
2. List pairs of factors of P that sum up to b.
3. Split bx as $px + qx$.
4. Factor by grouping terms in pairs:
$$ax^2 + px + qx + c = 0$$
5. Extract common binomial factor:
$$(px + r)(qx + s) = 0$$
6. Equate each linear factor to zero to find roots.

3.2 Factoring with Radical (Surd) Coefficients

TRAP ALERT

Handling Radicals in Middle and Constant Terms: When equations involve square roots like $\sqrt{k}$, calculate ac carefully:

- For $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$: Product $ac = (\sqrt{2})(5\sqrt{2}) = 5 \times 2 = \mathbf{10}$.
- Find two numbers whose product is 10 and sum is 7: 5 and 2 ($5 \times 2 = 10, 5 + 2 = 7$).
- Rewrite $2x$ strategically as $\sqrt{2} \cdot \sqrt{2}x$ to enable grouping!

Worked Archetype 3 | Factorization with Radical Coefficients

Problem: Solve the quadratic equation by factorization: $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$.

Step 1: Compute product ac and target sum b:

$$a = \sqrt{2}, \quad b = 7, \quad c = 5\sqrt{2} \implies ac = \sqrt{2} \times 5\sqrt{2} = 10$$

We need factors of 10 that add up to 7. The factors are 5 and 2.

Step 2: Split the middle term $7x$ into $5x + 2x$:

$$\sqrt{2}x^2 + 2x + 5x + 5\sqrt{2} = 0$$

Step 3: Group terms in pairs and factor out common elements: Notice that $2 = (\sqrt{2})^2$:

$$\sqrt{2}x(x + \sqrt{2}) + 5(x + \sqrt{2}) = 0$$

Step 4: Extract common binomial factor $(x + \sqrt{2})$:

$$(x + \sqrt{2})(\sqrt{2}x + 5) = 0$$

Step 5: Apply the Zero-Product Principle:

$$x + \sqrt{2} = 0 \implies \mathbf{x = -\sqrt{2}} \quad \text{or} \quad \sqrt{2}x + 5 = 0 \implies \mathbf{x = -\frac{5}{\sqrt{2}} = -\frac{5\sqrt{2}}{2}}$$

Final Statement: The roots of the equation are $\mathbf{-\sqrt{2}}$ and $\mathbf{-\frac{5\sqrt{2}}{2}}$.

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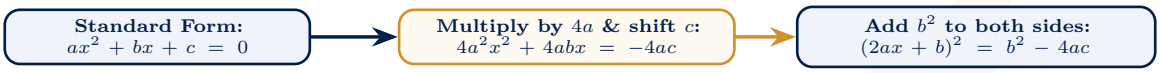
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QUADRATIC EQUATIONS

Module 4 | The Quadratic Formula: Sreedharacharya's Classical Derivation

4.1 Historical Context: Sreedharacharya's Elegant Technique

Ancient Indian mathematician **Sreedharacharya** (c. 8th–9th century CE) derived the universal quadratic formula using a brilliant device: instead of dividing by a (which introduces messy fractions), he **multiplied the entire equation by** $4a$. This creates an exact square term $(2ax)^2$ and $4abx$, which perfectly completes the binomial square $(2ax+b)^2$!



4.2 Rigorous Step-by-Step Proof of the Quadratic Formula

Algebraic Derivation of the Formula:

1. Start with the standard quadratic equation:

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

2. Transpose the constant term to RHS:

$$ax^2 + bx = -c$$

3. Multiply the entire equation by $4a$ (Sreedharacharya's device):

$$4a(ax^2 + bx) = 4a(-c) \implies 4a^2x^2 + 4abx = -4ac$$

4. Recognize $4a^2x^2 = (2ax)^2$ and $4abx = 2(2ax)(b)$. Add $\mathbf{b^2}$ to both sides:

$$(2ax)^2 + 2(2ax)(b) + b^2 = b^2 - 4ac$$

5. Factor the LHS as a perfect square:

$$(2ax + b)^2 = b^2 - 4ac$$

6. Take the square root of both sides (provided $b^2 - 4ac \geq 0$):

$$2ax + b = \pm\sqrt{b^2 - 4ac}$$

7. Solve for x by isolating $2ax$:

$$2ax = -b \pm \sqrt{b^2 - 4ac} \implies \mathbf{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

Worked Archetype 4 | Application of the Quadratic Formula

Problem: Solve the equation $2x^2 - 7x + 3 = 0$ using the quadratic formula.

Step 1: Identify coefficients: Comparing with $ax^2 + bx + c = 0 \implies a = 2,\ b = -7,\ c = 3$.

Step 2: Calculate the discriminant $D = b^2 - 4ac$:

$$D = (-7)^2 - 4(2)(3) = 49 - 24 = \mathbf{25} > \mathbf{0} \implies \text{Two distinct real roots exist.}$$

Step 3: Substitute into the quadratic formula:

$$x = \frac{-(-7) \pm \sqrt{25}}{2(2)} = \frac{7 \pm 5}{4}$$

Step 4: Separate the two roots:

$$x_1 = \frac{7+5}{4} = \frac{12}{4} = \mathbf{3}, \qquad x_2 = \frac{7-5}{4} = \frac{2}{4} = \mathbf{\frac{1}{2}}$$

Conclusion: The roots are **3** and $\frac{1}{2}$.

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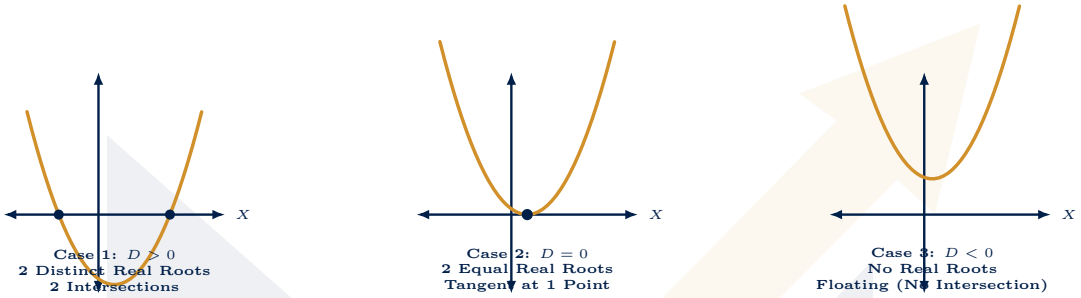
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Module 5 | The Discriminant (D) & Tri-State Nature of Roots Matrix

5.1 The Discriminant Function $D = b^2 - 4ac$

In the quadratic formula $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$, the quantity inside the square root, $\mathbf{D = b^2 - 4ac}$, is called the **discriminant** because it *discriminates* between the different types of roots.



5.2 The Comprehensive Tri-State Nature of Roots Table

Condition on D	Value of $\sqrt{D}$	Nature of Roots	Exact Formula for Roots
$\mathbf{D > 0}$	Real & Positive	**Two distinct real roots**	$x = \frac{-b \pm \sqrt{D}}{2a}$ (Rational if D is perfect square; conjugate surds if not).
$\mathbf{D = 0}$	Zero (0)	**Two equal real roots** (co-incident)	$\mathbf{x = -\frac{b}{2a}}$, $-\frac{\mathbf{b}}{\mathbf{2a}}$ (Parabola touches x-axis at vertex).
$\mathbf{D < 0}$	Imaginary	**No real roots**	No real values of x. Parabola does not touch x-axis.

Worked Archetype 5 | Inspecting the Nature of Roots

Problem: Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$ and hence find the nature of its roots.
Solution: Given $2x^2 - 4x + 3 = 0 \implies a = 2,\ b = -4,\ c = 3$.

$$D = b^2 - 4ac = (-4)^2 - 4(2)(3) = 16 - 24 = \mathbf{-8}$$

Since $\mathbf{D < 0}$, the given quadratic equation has **no real roots**.

Worked Archetype 6 | Coincident Roots ($D = 0$)

Problem: Find the nature of roots of $3x^2 - 2\sqrt{6}x + 2 = 0$. If real roots exist, find them.

$$D = b^2 - 4ac = (-2\sqrt{6})^2 - 4(3)(2) = [4 \times 6] - 24 = 24 - 24 = \mathbf{0}$$

Since $\mathbf{D = 0}$, the equation has **two equal real roots**:

$$x = -\frac{b}{2a} = -\frac{-2\sqrt{6}}{2(3)} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3} = \sqrt{\frac{\mathbf{2}}{\mathbf{3}}}$$

Hence, both roots are $\sqrt{\frac{\mathbf{2}}{\mathbf{3}}}$ and $\sqrt{\frac{\mathbf{2}}{\mathbf{3}}}$.

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Module 6 | The Parameter k Protocol: Condition for Equal Roots ($D = 0$)

6.1 The Tangency Condition and Parameter Determination

A quintessential question in the CBSE Class 10 board examination requires finding the value(s) of an unknown parameter k such that the quadratic equation has **two equal real roots** (or coincident roots).

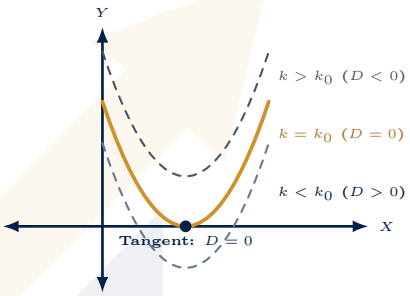
The Dual Criteria for Equal Roots:

1. **Criterion 1 (Discriminant Zero):**

$$\mathbf{D = 0 \iff b^2 - 4ac = 0 \iff b^2 = 4ac}$$

2. **Criterion 2 (Non-Zero Leading Coefficient):**

$$\mathbf{a \neq 0}$$



TRAP ALERT

The Fatal Parameter Trap: If the coefficient of x^2 contains k (e.g., $a = k - 12$), you **must explicitly reject** any value of k that makes $a = 0$!

6.2 Step-by-Step Equal Roots Solution Protocol

1. Expand all brackets and collect like terms to arrange the equation in strict standard form: $Ax^2 + Bx + C = 0$.
2. Write down the expressions for coefficients a, b, c in terms of k.
3. Set up the discriminant equation: $D = b^2 - 4ac = 0$.
4. Solve the resulting algebraic equation for k.
5. **Mandatory Check:** Ensure that the values of k do not cause the leading coefficient a to become zero.

Worked Archetype 7 | The Classic Leading Coefficient Trap

Problem: Find the value(s) of k for which the quadratic equation $(k-12)x^2 + 2(k-12)x + 2 = 0$ has two equal real roots.

Step 1: Identify coefficients:

$$a = k - 12, \quad b = 2(k-12), \quad c = 2$$

Constraint: For the equation to remain quadratic, $a \neq 0 \implies k - 12 \neq 0 \implies \mathbf{k \neq 12}$.

Step 2: Apply the equal roots condition $D = 0$:

$$D = b^2 - 4ac = [2(k-12)]^2 - 4(k-12)(2) = 0$$

$$4(k-12)^2 - 8(k-12) = 0$$

Step 3: Factor out the common term $4(k-12)$ (DO NOT divide by $k-12$ blindly!):

$$4(k-12)\big[(k-12) - 2\big] = 0$$

$$4(k-12)(k-14) = 0$$

This yields two algebraic solutions:

$$k - 12 = 0 \implies k = 12 \quad \text{or} \quad k - 14 = 0 \implies k = 14$$

Step 4: Check domain validity: If $k = 12$, then $a = 12 - 12 = 0$, so the equation collapses to $0 \cdot x^2 + 0 \cdot x + 2 = 0 \iff 2 = 0$ (impossible!).

Hence, $\mathbf{k = 12}$ is **extraneous and must be rejected**.

Conclusion: The only valid value of k is $\mathbf{k = 14}$.

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Module 7 | Word Problems I: Numbers, Reciprocals & Age Models

7.1 Translation Protocol from Natural Language to Quadratic Equations

Word problems test your ability to translate verbal relationships into standard quadratic equations:

Verbal Problem Model	Algebraic Representation	Standard Form Equation
Two consecutive integers	x and $x+1$	Product $= P \implies x(x+1) = P \implies x^2 + x - P = 0$
Two consecutive odd/even integers	x and $x+2$	Sum of squares $= S \implies x^2 + (x+2)^2 = S$
A number and its reciprocal	x and $\frac{1}{x}$ $(x \neq 0)$	$x + \frac{1}{x} = \frac{p}{q} \implies qx^2 - px + q = 0$
Two numbers with sum S, product P	x and $S - x$	$x(S-x) = P \implies x^2 - Sx + P = 0$
Age problems (years ago / hence)	Present age $= x$; Past $= x - n$; Future $= x + m$	Product of ages $= P \implies (x-n)(x+m) = P$

7.2 Domain Validation: Rejecting Non-Physical Solutions

TRAP ALERT

Physical Variables Have Strict Domain Restrictions: Quadratic equations mathematically yield two roots. In word problems, you must test both roots against physical reality:

- **Ages:** Must be positive real numbers ($x > 0$). Reject negative roots!
- **Natural Numbers / Counts:** Must be positive integers ($x \in \mathbb{N}$).
- **Dimensions (Length, Width):** Must be strictly positive ($x > 0$).

Worked Archetype 8 | The Reciprocal Age Problem

Problem: The sum of the reciprocals of Rehman's ages (in years) 3 years ago and 5 years from now is $\frac{1}{3}$. Find his present age.

Step 1: Define the variable: Let Rehman's present age be x years ($x > 3$).
Rehman's age 3 years ago $= (x - 3)$ years.
Rehman's age 5 years from now $= (x + 5)$ years.

Step 2: Formulate the equation according to given conditions:

$$\frac{1}{x-3} + \frac{1}{x+5} = \frac{1}{3}$$

Step 3: Combine fractions by taking LCM on LHS:

$$\frac{(x+5)+(x-3)}{(x-3)(x+5)} = \frac{1}{3} \implies \frac{2x+2}{x^2+2x-15} = \frac{1}{3}$$

Step 4: Cross-multiply and simplify to standard form:

$$3(2x+2) = x^2 + 2x - 15 \implies 6x + 6 = x^2 + 2x - 15$$

$$x^2 + 2x - 6x - 15 - 6 = 0 \implies \mathbf{x^2 - 4x - 21 = 0}$$

Step 5: Factor by splitting the middle term ($ac = -21, b = -4 \implies -7$ and 3):

$$(x-7)(x+3) = 0 \implies x = 7 \quad \text{or} \quad x = -3$$

Step 6: Physical validation: Age cannot be negative, so $x = -3$ is rejected.
Conclusion: Rehman's present age is **7** years.

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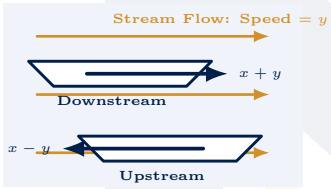
Module 8 | Word Problems II: Speed, Distance & River Kinematics

8.1 The Speed-Distance-Time Identity: $t = \frac{d}{s}$

All uniform motion problems rely on the relation Time $= \frac{\text{Distance}}{\text{Speed}}$. When speed changes, a time differential Δt is created between the slow and fast trips:

$$\mathbf{t_{slow} - t_{fast} = \Delta t \implies \frac{d}{s} - \frac{d}{s+v} = \Delta t}$$

8.2 River Stream Kinematics (Upstream vs. Downstream)



Relative Velocities on Water: Let boat speed in still water be x km/h, and stream speed be y km/h ($x > y$):

- **Downstream (with flow):**

$$\mathbf{v_{down} = x + y}$$

- **Upstream (against flow):**

$$\mathbf{v_{up} = x - y}$$

- **Time Equation for Distance d:** Since $x - y < x + y$, upstream travel takes **longer**:

$$\mathbf{t_{up} - t_{down} = \Delta t \iff \frac{d}{x-y} - \frac{d}{x+y} = \Delta t}$$

Worked Archetype 9 | The Motorboat Stream Problem

Problem: A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

Step 1: Set up variables: Let the speed of the stream be y km/h ($0 < y < 18$).
Speed upstream $= (18 - y)$ km/h, Speed downstream $= (18 + y)$ km/h.

Step 2: Formulate the time equation ($t_{up} - t_{down} = 1$ hour):

$$\frac{24}{18-y} - \frac{24}{18+y} = 1$$

Step 3: Factor out 24 and combine fractions:

$$24\left[\frac{(18+y)-(18-y)}{(18-y)(18+y)}\right] = 1 \implies 24\left[\frac{2y}{324-y^2}\right] = 1$$

$$\frac{48y}{324-y^2} = 1 \implies 48y = 324 - y^2 \implies \mathbf{y^2 + 48y - 324 = 0}$$

Step 4: Factor the quadratic equation ($ac = -324, b = 48 \implies 54$ and -6):

$$(y+54)(y-6) = 0 \implies y = -54 \quad \text{or} \quad y = 6$$

Step 5: Physical validation: Speed cannot be negative $\implies y = -54$ is rejected.
Conclusion: The speed of the stream is **6** km/h.

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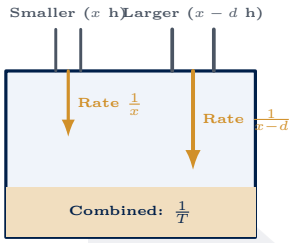
CORE SYLLABUS

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Module 9 | Word Problems III: Water Taps, Cisterns & Unit Work Rates

10.1 The Unit Work Rate Principle

If a pipe or worker can complete a task in T hours, then in 1 hour, the fraction of work completed is exactly $\frac{1}{T}$. When two pipes work simultaneously, their individual flow rates add up to equal the combined flow rate:



Master Tap Flow Rate Equation: Let smaller tap take x hours, and larger tap take $(x-d)$ hours to fill tank alone:

$$\text{Rate of Tap 1} = \frac{\mathbf{1}}{\mathbf{x}}$$

$$\text{Rate of Tap 2} = \frac{\mathbf{1}}{\mathbf{x-d}}$$

$$\text{Combined Rate} = \frac{\mathbf{1}}{\mathbf{x}} + \frac{\mathbf{1}}{\mathbf{x-d}} = \frac{\mathbf{1}}{\mathbf{T}_{\text{together}}}$$

Clearing denominators produces a clean quadratic equation in x!

Worked Archetype 10 | Two Water Taps Filling a Tank Together

Problem: Two water taps together can fill a tank in $9\frac{3}{8}$ hours ($= \frac{75}{8}$ hours). The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.

Step 1: Define variables: Let time taken by smaller tap alone $= x$ hours ($x > 10$).
Time taken by larger tap alone $= (x-10)$ hours.
Combined time $T = \frac{75}{8}$ hours $\implies$ Combined 1-hour rate $= \frac{1}{T} = \frac{8}{75}$.

Step 2: Set up the flow rate equation:

$$\frac{1}{x} + \frac{1}{x-10} = \frac{8}{75}$$

Step 3: Combine LHS and cross-multiply:

$$\frac{(x-10)+x}{x(x-10)} = \frac{8}{75} \implies \frac{2x-10}{x^2-10x} = \frac{8}{75}$$

Divide numerator of LHS and RHS by 2:

$$\frac{x-5}{x^2-10x} = \frac{4}{75} \implies 75(x-5) = 4(x^2-10x)$$

$$75x - 375 = 4x^2 - 40x \implies \mathbf{4x^2 - 115x + 375 = 0}$$

Step 4: Solve the quadratic equation by factoring ($ac = 4 \times 375 = 1500, b = -115 \implies -100$ and -15):

$$4x^2 - 100x - 15x + 375 = 0 \implies 4x(x-25) - 15(x-25) = 0$$

$$(x-25)(4x-15) = 0 \implies x = 25 \quad \text{or} \quad x = \frac{15}{4} = 3.75$$

Step 5: Physical validity check: If $x = 3.75$ hours, then time for larger tap $= 3.75 - 10 = -6.25$ hours (impossible!).
Hence, $x = 3.75$ is **rejected**, and we accept $\mathbf{x = 25}$.
Conclusion: Time for smaller tap alone is **25** hours, and for larger tap alone is **15** hours.

QUADRATIC EQUATIONS

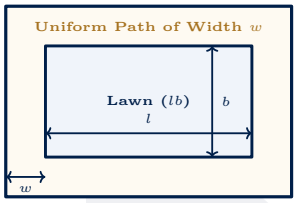
Module 10 | Word Problems IV: Geometry, Mensuration & Pathways

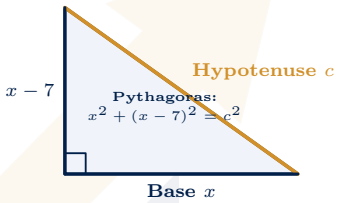
11.1 Geometric Formulations via Pythagoras & Area Theorems

Mensuration word problems translate spatial configurations into quadratic forms:

- **Right Triangles:** Hypotenuse c, sides x and $x-d \implies \mathbf{x^2 + (x-d)^2 = c^2}$.
- **Rectangular Fields with Uniform Borders:** An inner lawn $(l \times b)$ surrounded by a path of width w:

$$\text{Outer Length} = l + 2w, \quad \text{Outer Breadth} = b + 2w \implies \text{Area of Path} = \mathbf{(l+2w)(b+2w) - lb}$$





Worked Archetype 11 | Right Triangle Altitude and Base Relationship

Problem: The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.

Step 1: Formulate variables: Let base $= x$ cm $(x > 7)$. Altitude $= (x-7)$ cm. Hypotenuse $= 13$ cm.

Step 2: Apply Pythagoras Theorem:

$$\text{Base}^2 + \text{Altitude}^2 = \text{Hypotenuse}^2 \implies x^2 + (x-7)^2 = 13^2$$

Step 3: Expand and simplify to standard form:

$$x^2 + (x^2 - 14x + 49) = 169 \implies 2x^2 - 14x + 49 - 169 = 0$$

$$2x^2 - 14x - 120 = 0 \implies \text{Divide by 2: } \mathbf{x^2 - 7x - 60 = 0}$$

Step 4: Factor the quadratic equation $(ac = -60, b = -7 \implies -12$ **and** $5)$**:**

$$(x-12)(x+5) = 0 \implies x = 12 \quad \text{or} \quad x = -5$$

Step 5: Physical validation: Side length cannot be negative $\implies x = -5$ is rejected.

$$\text{Base } x = \mathbf{12} \text{ cm}, \quad \text{Altitude } = x - 7 = 12 - 7 = \mathbf{5} \text{ cm}$$

Verification: $12^2 + 5^2 = 144 + 25 = 169 = 13^2$ (Pythagorean triplet verified!).

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CHAPTER 04

QUICK REFERENCE

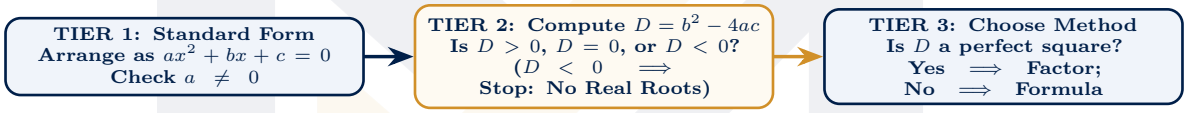
QUADRATIC EQUATIONS

Module 11 | Master Formula Matrix & Solution Decision Protocol

11.1 The Comprehensive Quadratic Equations Matrix

Method / Theorem	Prerequisite Form	Mathematical Rule	Strategic Board Examination Purpose
Standard Form	$a \neq 0$	$ax^2 + bx + c = 0$	Mandatory first step for all quadratic methods.
Factorization	$p \cdot q = ac,\ p + q = b$	$(px + r)(qx + s) = 0$	Fastest method when ac factors cleanly.
Quadratic Formula	$D \geq 0$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Universal method; handles surd/radical coefficients.
Discriminant D	$D = b^2 - 4ac$	$D > 0,\ D = 0,\ D < 0$	Testing nature of roots without solving equation.
Equal Roots Condition	Two equal roots	$b^2 - 4ac = 0$ with $a \neq 0$	Finding unknown parameter k (high frequency!).
Parabola Vertex	Axis of symmetry	$x = -\frac{b}{2a},\ y = -\frac{D}{4a}$	Extreme values / trajectory height peak.
Sum & Product	Roots α, β	$\alpha + \beta = -\frac{b}{a},\ \alpha\beta = \frac{c}{a}$	Fast verification check of computed roots.

11.2 The 3-Tier Board Problem Decision Flowchart



Fast Mental-Math Factoring Rules:

- **Sign of c is Positive ($c > 0$):** Both factors must have the **same sign** (the sign of b).
  - $x^2 - 8x + 12 = 0 \implies b < 0, c > 0 \implies$ both negative: $(x - 6)(x - 2) = 0$.
  - $x^2 + 8x + 12 = 0 \implies b > 0, c > 0 \implies$ both positive: $(x + 6)(x + 2) = 0$.
- **Sign of c is Negative ($c < 0$):** Factors have **opposite signs**; the larger factor takes the sign of b.
  - $x^2 - x - 12 = 0 \implies b < 0, c < 0 \implies$ larger factor is negative: $(x - 4)(x + 3) = 0$.

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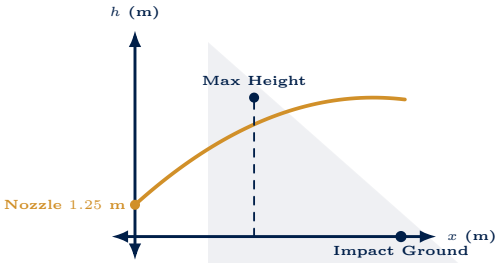
QUADRATIC EQUATIONS

Module 12 | Applied Case-Study: Projectile Motion & Parabolic Flight

Case Study 1 | Fire Service Water Cannon Trajectory

A specialized fire-fighting truck sprays water through a pressurized cannon. The height h (in meters) of the water jet above the ground at a horizontal distance x (in meters) from the nozzle is modeled by the quadratic equation:

$$h(x) = -\frac{1}{20}x^2 + x + 1.25$$



4-Mark Board Question Protocol

Questions:

1. What is the height of the nozzle from the ground? [1M]
2. At what horizontal distance does the water jet reach its maximum height? [1M]
3. Find the horizontal distance from the nozzle where the water jet strikes the ground. [2M]

Solution & Marking Scheme Evaluation

1. **Height of Nozzle [1M]:** At the nozzle, the horizontal distance is $x = 0$:

$$h(0) = -\frac{1}{20}(0)^2 + (0) + 1.25 = \mathbf{1.25} \text{ meters}$$

2. **Distance for Maximum Height [1M]:** Maximum height occurs at the vertex $x = -\frac{b}{2a}$:

$$a = -\frac{1}{20} = -0.05, \quad b = 1 \implies x = -\frac{1}{2(-0.05)} = -\frac{1}{-0.1} = \mathbf{10} \text{ meters}$$

3. **Impact Horizontal Distance [2M]:** Water strikes ground when height $h(x) = 0$:

$$-\frac{1}{20}x^2 + x + 1.25 = 0 \implies \text{Multiply by } -20: \quad \mathbf{x^2 - 20x - 25 = 0}$$

Apply the quadratic formula with $a = 1, b = -20, c = -25$:

$$D = (-20)^2 - 4(1)(-25) = 400 + 100 = 500$$

$$x = \frac{-(-20) \pm \sqrt{500}}{2(1)} = \frac{20 \pm 10\sqrt{5}}{2} = 10 \pm 5\sqrt{5}$$

Since distance must be positive, take the positive root ($\sqrt{5} \approx 2.236$):

$$x = 10 + 5(2.236) = 10 + 11.18 = \mathbf{21.18} \text{ meters}$$

CHAPTER 04

EXAM STRATEGY

QUADRATIC EQUATIONS

Module 13 | Board Traps, Verification Hacks & 5-Point Audit

13.1 The Top 7 Quadratic Equation Board Traps (And How to Bust Them)

Trap 1: The Variable Division Cancellation Trap: Solving $x^2 = 5x$ by dividing both sides by x to get $x = 5$.

TRAP ALERT

You lose the valid root $\mathbf{x = 0}$! Always factor: $x^2 - 5x = 0 \implies x(x-5) = 0 \implies x = 0, 5$.

Trap 2: The Negative Linear Coefficient Sign Error: In $x = \frac{-b \pm \sqrt{D}}{2a}$, when $b = -7$, writing $-7 \pm \ldots$ instead of $-(-7) = +\mathbf{7}$.

Trap 3: The Squaring Negative b Blunder: Writing $(-6)^2 = -36$ inside $b^2 - 4ac$. Any real number squared is non-negative: $\mathbf{(-6)^2 = +36}$!

Trap 4: The Parameter $a \neq 0$ Violation: In $(k-1)x^2 + 2x + 1 = 0$, finding $k = 1$ from $D = 0$ but forgetting that $k = 1$ makes $a = 0$. You **must reject** values making the leading coefficient zero!

Trap 5: Upstream vs. Downstream Subtraction Order: Writing downstream time minus upstream time as positive. Upstream is always **slower**, so $\mathbf{t}_{\text{upstream}} > \mathbf{t}_{\text{downstream}}$.

Trap 6: Retaining Extraneous Negative Roots in Word Problems: Forgetting to state: "Since speed/age/time cannot be negative, $x = -k$ is rejected."

Trap 7: Using the Deleted "Completing the Square" Method: In CBSE 2026–27, solve strictly using **Factorization** or the **Quadratic Formula**.

13.2 High-Speed Algebraic Verification Protocol (Vieta's Relations)

After calculating roots α and β, check them mentally in 5 seconds using Vieta's formulas:

$$\text{Sum of Roots: } \alpha + \beta = -\frac{\mathbf{b}}{\mathbf{a}} \qquad \text{and} \qquad \text{Product of Roots: } \alpha \cdot \beta = \frac{\mathbf{c}}{\mathbf{a}}$$

If the sum or product does not match the original equation coefficients, an arithmetic error occurred!

13.3 The 5-Point Board Examination Pre-Submission Checklist

Final 60-Second Answer Sheet Verification Protocol:

- ☐ **Standard Form Established:** Did you gather all terms to one side ($ax^2 + bx + c = 0$) before identifying a, b, c?
- ☐ **Discriminant Sign Verified:** Did you compute $D = b^2 - 4ac$ accurately with correct sign multiplication $(-4ac)$?
- ☐ **Two Roots Recorded:** Did you state both roots explicitly? (Quadratic equations have 2 roots!).
- ☐ **Physical Feasibility Justified:** For word problems, did you write an explicit sentence rejecting the negative root?
- ☐ **Correct Dimensional Units Attached:** Speeds in km/h, Times in hours, Dimensions in cm/m!

MATHS MANTHAN TUITION CLASSES | END OF CHAPTER 4 HANDBOOK

Designed for Academic Excellence in CBSE Class 10 Board Examinations 2026–27.